Digital Forensics 1 Rubric 4

Network Forensics Examination
The project must include capture and examination of network traffic using Wireshark with proper filtering. Analyze system/network logs for evidence using LogParser or Splunk. Collect network evidence including browser history and connection logs. Create a timeline of network activity. Documentation must include packet analysis, significant events, artifact extraction, and timestamp correlation with supporting evidence.

To satisfy the **Network Forensics Examination** section of your rubric, your report must demonstrate the ability to capture, filter, and analyze traffic moving across the wire, and correlate it directly with host-based network artifacts and centralized logs.

The core requirement here is packet-to-host correlation: proving that an action seen inside a Wireshark packet capture (pcap) matches a local browser history entry, an active network connection, and a centralized system log at the exact same timestamp.

Here is how to execute and document these requirements to achieve maximum rubric points.

🌐 1. Wireshark Packet Analysis & Filtering

You must capture or ingest a network packet capture (`.pcap`/`.pcapng`) and use precise display filters to isolate malicious or unauthorized activity.

- **Key Filters to Execute & Document:**
    - **Isolating the Host:** `ip.addr == [Target_IP]`
    - **Web Traffic / Data Exfiltration:** `http.request` or `tls.handshake.type == 1` (to find domains visited via HTTPS SNI).
    - **Unusual Protocols / Port Scans:** `tcp.flags.syn == 1 and tcp.flags.ack == 0` (to find port scanning) or filtering for non-standard ports.
    - **DNS Reconnaissance:** `dns.flags.response == 0` to look at what domains the suspect system was resolving.

- **Artifact Extraction via Wireshark:**
    - If unencrypted HTTP traffic was used, show the extraction of files: Go to **File** → **Export Objects** → **HTTP**, and save the file.
- **Documentation Requirement:** Include a screenshot of Wireshark showing your specific filter applied in the filter bar, with the relevant malicious packet details expanded in the *Packet Details* pane.

📊 2. Log Analysis using LogParser or Splunk

You must ingest system, firewalls, or network device logs into **Splunk** or query them using **LogParser** to identify anomalies.

- **LogParser Execution (Example for Windows IIS Web Logs or Event Logs):**
    - Run a query to pull failed connections or specific IP interactions:
      ```
      LogParser.exe -i:EVT "SELECT TimeGenerated, EventID, SourceName, Message FROM Security WHERE EventID=4624 AND Message LIKE '%Network%'" -o:DATAGRID
      ```
- **Splunk Execution:**
    - Query your ingested network/host logs using SPL (Splunk Processing Language):
      ```
      index=network_logs src_ip="[Target_IP]" status=404 | table _time, dest_ip, uri_path
      ```
- **Documentation Requirement:** Take a screenshot of your LogParser command/grid or your Splunk search bar showing the query results. Document the exact parameters used.

🌐 3. Host-Based Network Artifacts (Browser History & Connections)

Network evidence doesn't just live on the wire; it leaves traces on the local storage. You must pull host-based connection logs and browser records.

- **Live Connection Logs:** Document active connections during the live capture phase using the command line: `netstat -ano` or `Get-NetTCPConnection`. This links a remote IP address to a specific local Process ID (PID).
- **Browser History & Cache:** Use tools inside **Autopsy** (Web History module) or standalone utilities like **BrowsingHistoryView** or **ChromeCacheView** to extract the local history database (`History` SQLite file).
- **Correlation:** Prove that the time a user visited a URL in their browser history matches the exact timestamp of the DNS request and TCP handshake found in your Wireshark pcap.

⏰ 4. Timeline of Network Activity

Synthesize your packet analysis, log queries, and host artifacts into a highly detailed **Network Timeline**. Every event must be correlated across multiple data sources to establish undisputed facts.

Timestamp (UTC)	Source / Layer	Event / Protocol	Activity Description	Cross-Evidence Correlation (The "Proof")
14:22:05	Host (`netstat`)	TCP Conn	Local PID 4322 establishes connection to remote IP `192.168.1.50:443`.	*Matches active connection state caught during live triage.*
14:22:06	Wireshark Pcap	TLS Handshake	Client Hello specifies Server Name Indication (SNI): `unauthorized-storage.com` .	*Packet #4,521. Confirms the destination domain of the PID 4322 connection.*

14:22:15	Autopsy Extraction	Chrome History	User navigates to `https://unauthorized-storage.com`.	*SQLite DB entry matches the destination IP and TLS timestamp precisely.*
14:25:30	Splunk (Firewall)	Network Log	Outbound data spike of 450MB detected from `Target_IP` to `192.168.1.50`.	*Log Entry ID 99423. Confirms large data exfiltration event took place during session.*

📝 Step-by-Step Reporting Layout for Rubric 4

To ensure the evaluator can easily check off every grading item, structure this section of your report as follows:

1. **Network Evidence Acquisition:** State the versions of Wireshark and LogParser/Splunk used. Document the network interface or log directory targeted.
2. **Packet Analysis & Filtering Logic:** Explain the exact display filters used and *why* they were chosen to isolate the traffic.
3. **Significant Events & Artifact Extraction:** Detail any files extracted from the network or specific malicious domains contacted.
4. **Timestamp Correlation & Timeline:** Insert the synchronized network timeline table showing host, network, and server log alignments.